

Input:-

variable = input ()

↳ taking input from the user

Band Name Generator :

name

your best pet

your best food.

Ex-

Azaan

Gini

Meat

→

Azaan Gini Meat

↳ wearing on my hand

Operators:-

$x + y$
 $x - y$
 $x * y$
 x / y

→ operators

→ asterisk (present on 8 number)

Python operators is like the operator we are having in mathematics like addition, subtraction, multiplication and dividation. Similary, these are used for doing mathematical calculation in python.

Types of Operators :

1. Arithmetic Operators
2. Assignment Operators
3. Relational Operators
4. Logical Operators
5. Bitwise Operators ?
6. Identity Operators ?

1. Arithmetic Operators

$+$ $\rightarrow$ Addition

$-$ $\rightarrow$ Subtraction

$*$ $\rightarrow$ Multiplication

$/$ $\rightarrow$ division

$//$ $\rightarrow$ floor division

$\%$ $\rightarrow$ modulus operator

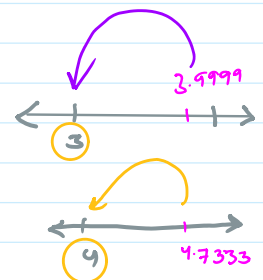
$**$ $\rightarrow$ Exponentiation left most

round off value

$10 // 3 = 3$

$10 / 3 = 3.33$

$\rightarrow$ nearest round off value



modulus ($\%$) :-

$\rightarrow$ It gives remainder as result

Ex:- $10 \% 3 = 1$

$10 \% 5 = 0$

$$\begin{array}{r} 3 \\ 3 \overline{) 10} \\ \underline{- 9} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \\ 5 \overline{) 10} \\ \underline{- 10} \\ 0 \end{array}$$

Note:- $a \% b$

If $a < b$

$\downarrow$

Ans = a

↓
ans = a
↖
2/1.5 = 2

Exponentiation :-

$$2^2 = 4 \Rightarrow 2 * 2$$

$$2^3 = 8 \Rightarrow 2 * 2 * 2$$

$$3^3 = 27 \Rightarrow 3 * 3 * 3$$

$$10^3 = 1000 \Rightarrow 10 * 10 * 10$$

2. Assignment Operators → It is combination of arithmetic & equals operators

+=

-=

*=

/=

//=

%=

**=

x = 10

x += 1

print(x) O/P ⇒ 11

x += 1 → x = x + 1

x = 10 + 1 ⇒ 11

x = 10

x -= 5

print(x) O/P ⇒ 5

x -= 5 → x = x - 5

x = 10 - 5 = 5

3. Relational Operators → Output of this either TRUE OR FALSE

== → equals to

>

<

x = 10 y = 10 x == y → True

x = 10 y = 100 x == y → False

$<$
 $>$
 $\leq$
 $\geq$
 $!=$

$x=10$ $y=10$ $x==y \rightarrow \text{true}$
 $x=2$ $y=10$ $x==y \rightarrow \text{false}$

$a=10$ $b=11$
 $a > b \rightarrow 10 > 11 \times \text{false}$

$x=10$ $y=2$
 $x > 2 \rightarrow x > 2 \text{ or } x == 2$
 $\downarrow$
 True

$!=$ $\rightarrow$ not equals $\rightarrow$ give True when values are not equal otherwise False.
 $x=10$ $y=5$
 $x != y \rightarrow \text{True}$

$x=7$
 $x \geq 2 \rightarrow \text{True}$

$x=2$
 $x \geq 2$
 $\downarrow$
 True

$x=10$ $y=10$
 $x != y \rightarrow \text{false}$

4. Logical Operators

AND
 OR

both are true $\rightarrow$ then TRUE
 $x > 2$ and $x > 10$
 $x > 2$ OR $x > 10$
 either one is true $\rightarrow$ then true

Home Work :

1. Armstrong Number : [Sandbox: A Kid-Friendly Platform for Coding & Math | Codeyoung](#)
2. Even or Odd : [Sandbox: A Kid-Friendly Platform for Coding & Math | Codeyoung](#)